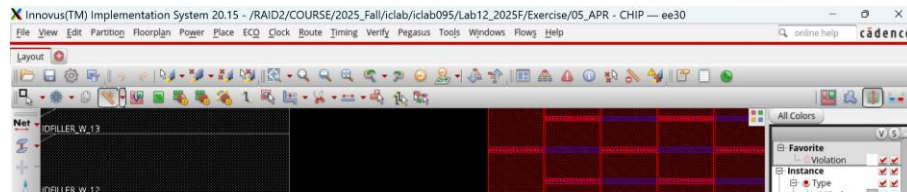


Report

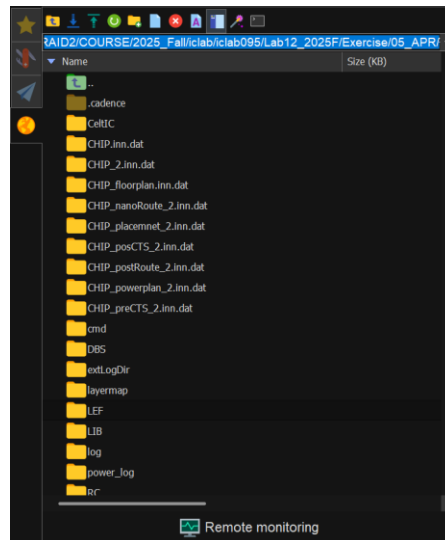
1. server info



2. server info

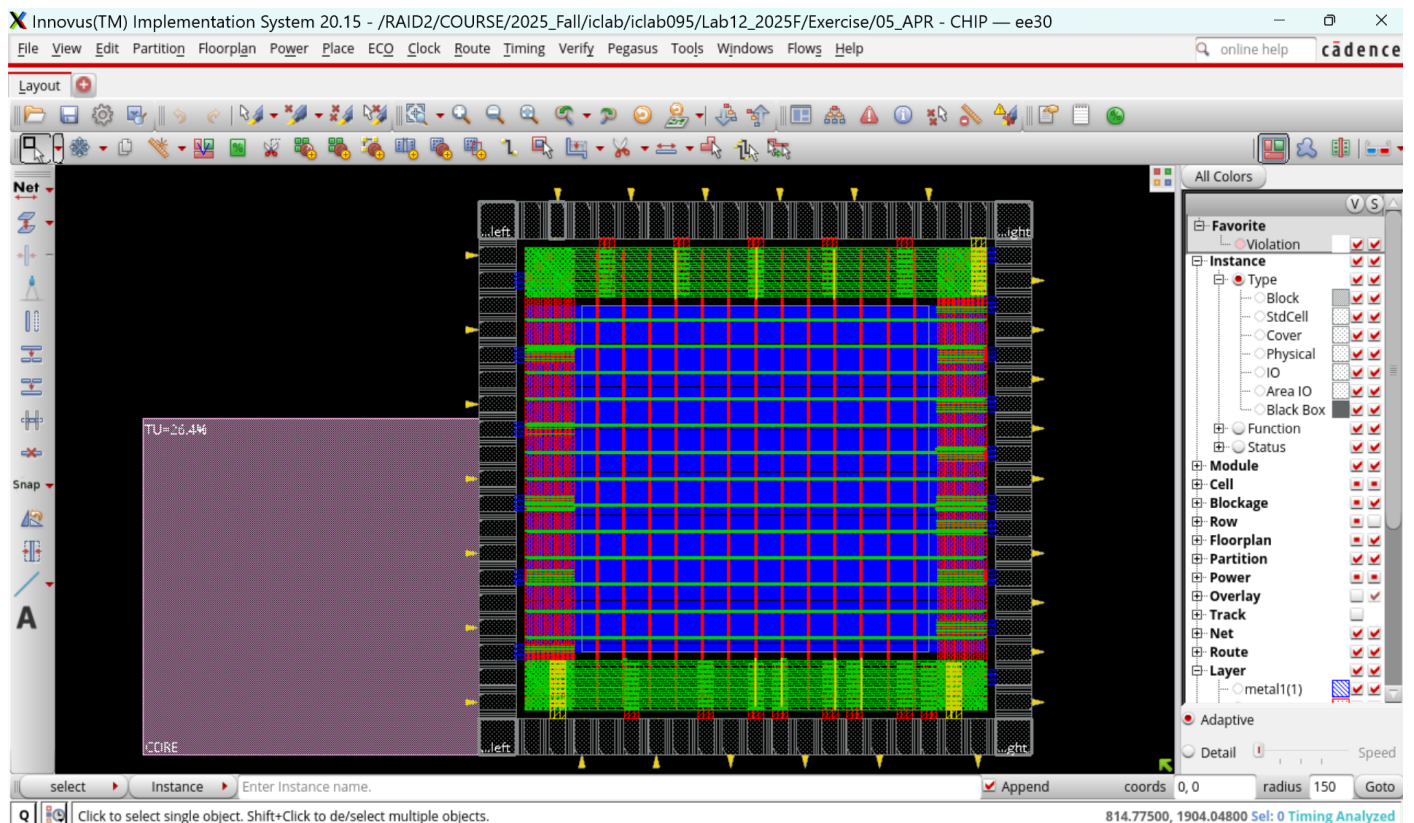
```
21:16 iclab095@ee30[~/Lab12_2025F/Exercise]$ cd 05_APR/  
21:16 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ setenv OA_HOME /RAID2/cad/cadence/INNOVUS  
/INNOVUS_20.15.000/share/oa  
21:17 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ innovus
```

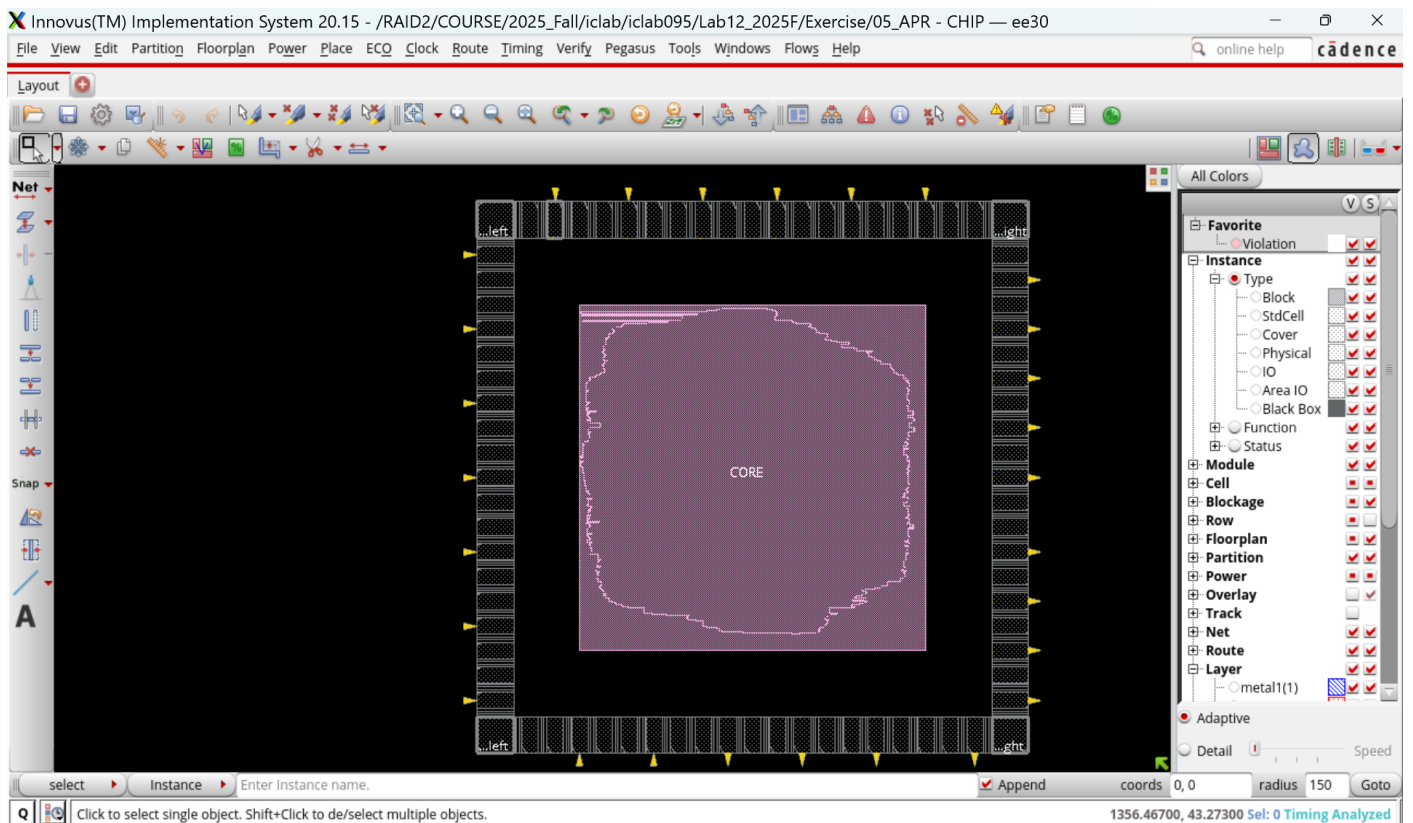
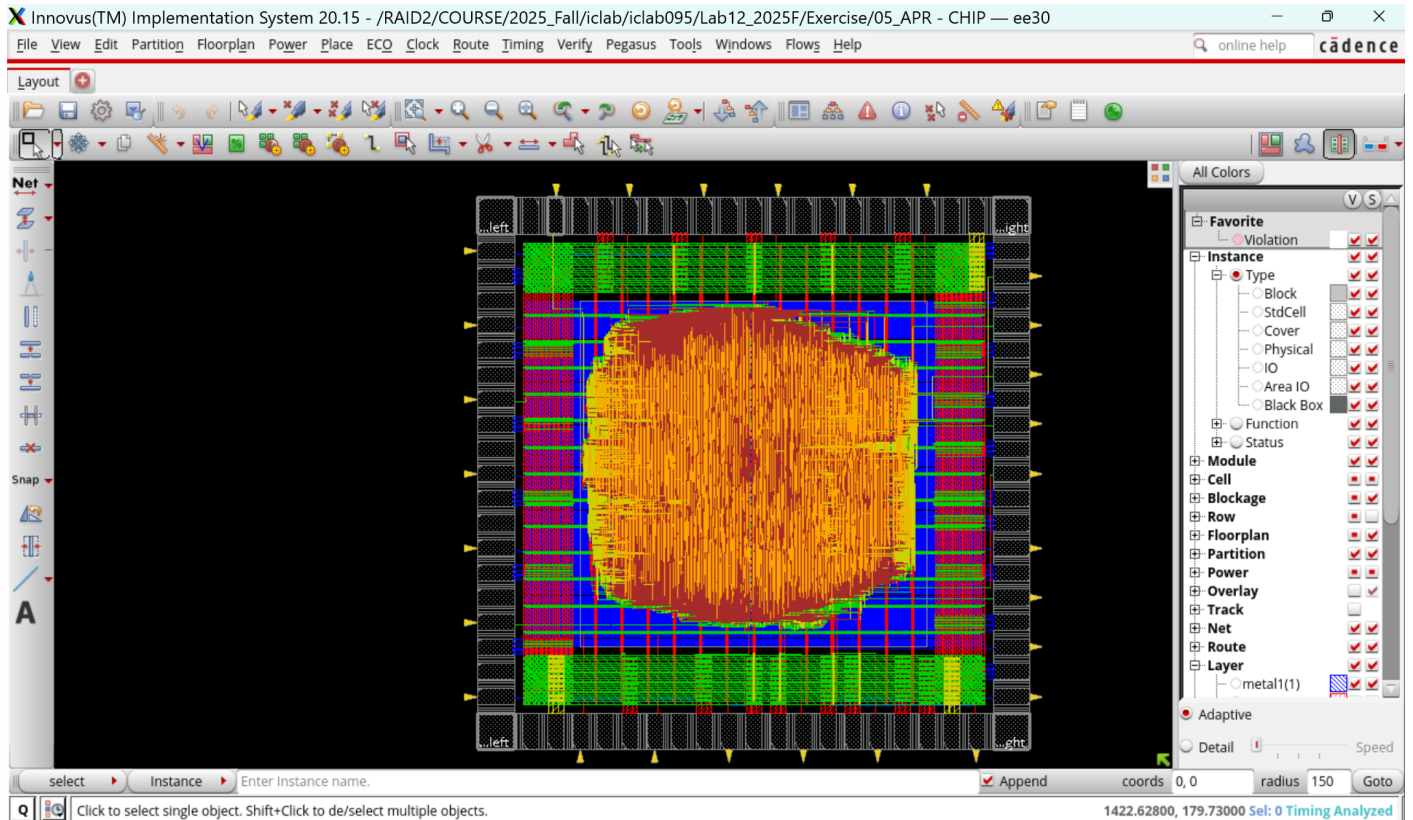
3. server info



1. Chip Layout View

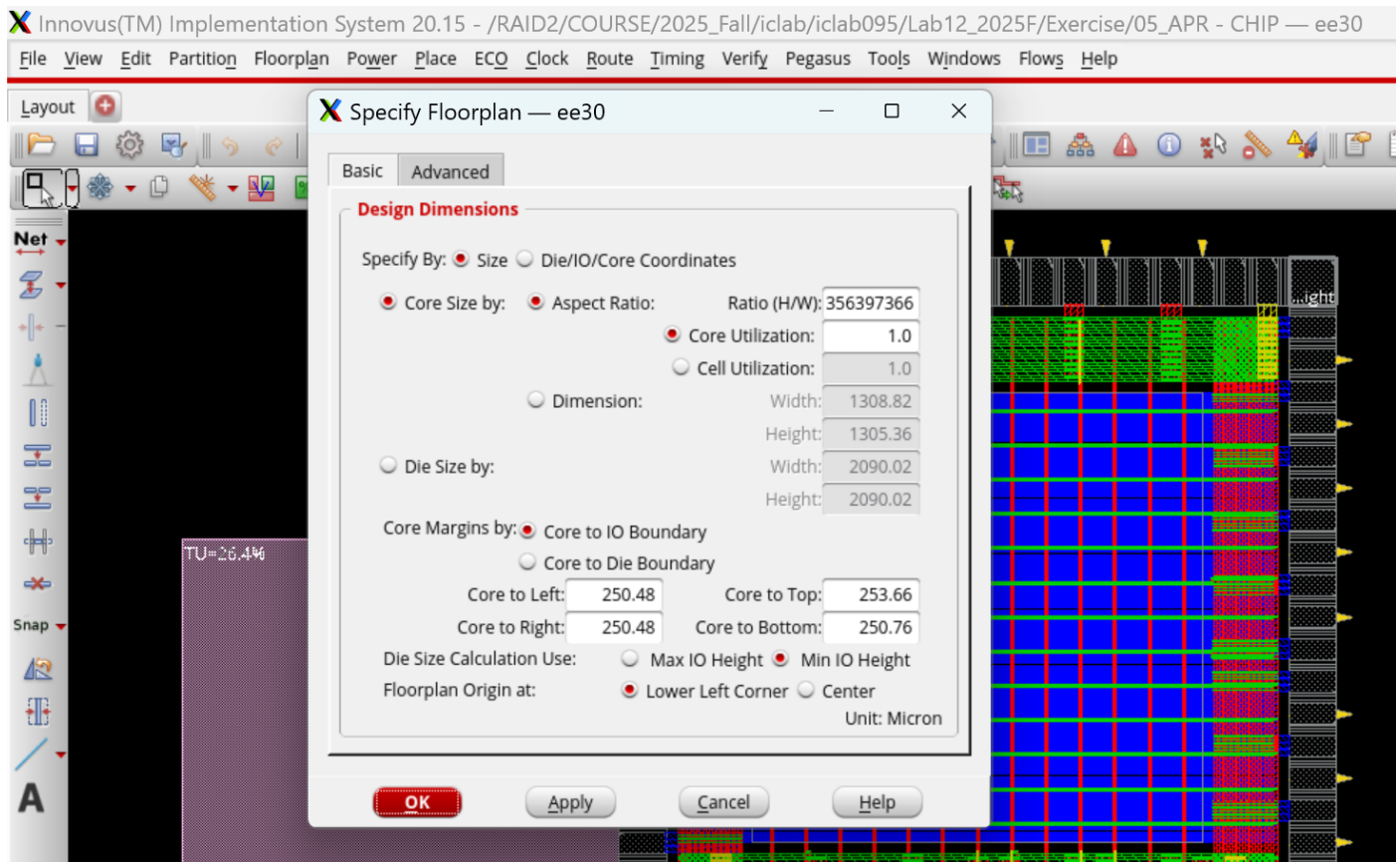
Instructions: After completing the final layout, capture a screenshot of the entire chip layout.





2. Core to IO boundary

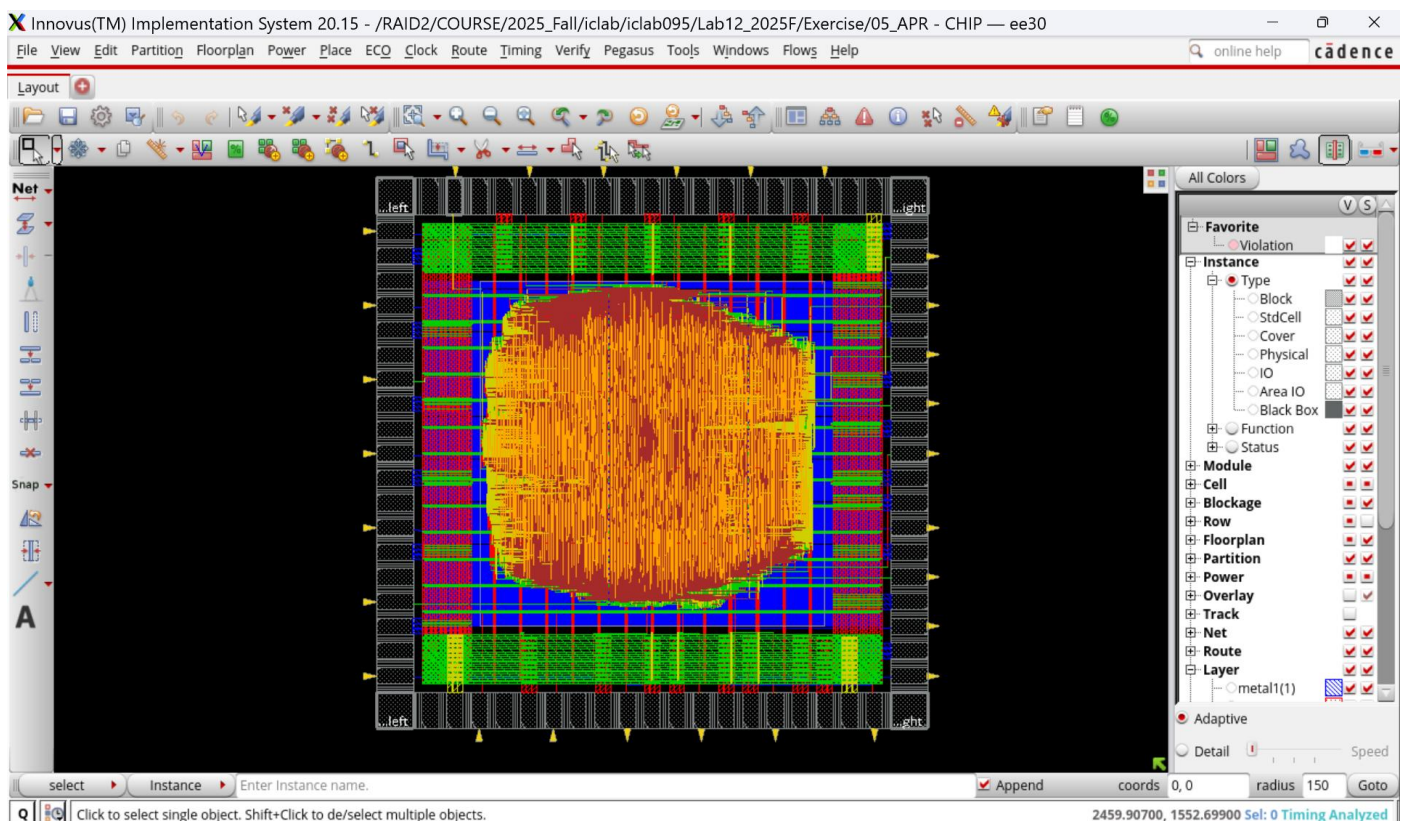
Instructions: After completing the final layout, open Floorplan -> Specify Floorplan, and capture a screenshot.



3. Core Ring

Instructions: After completing the final layout, zoom in on the Power section (ensuring the entire power ring is visible), and use the K key to measure the width of a single power line.

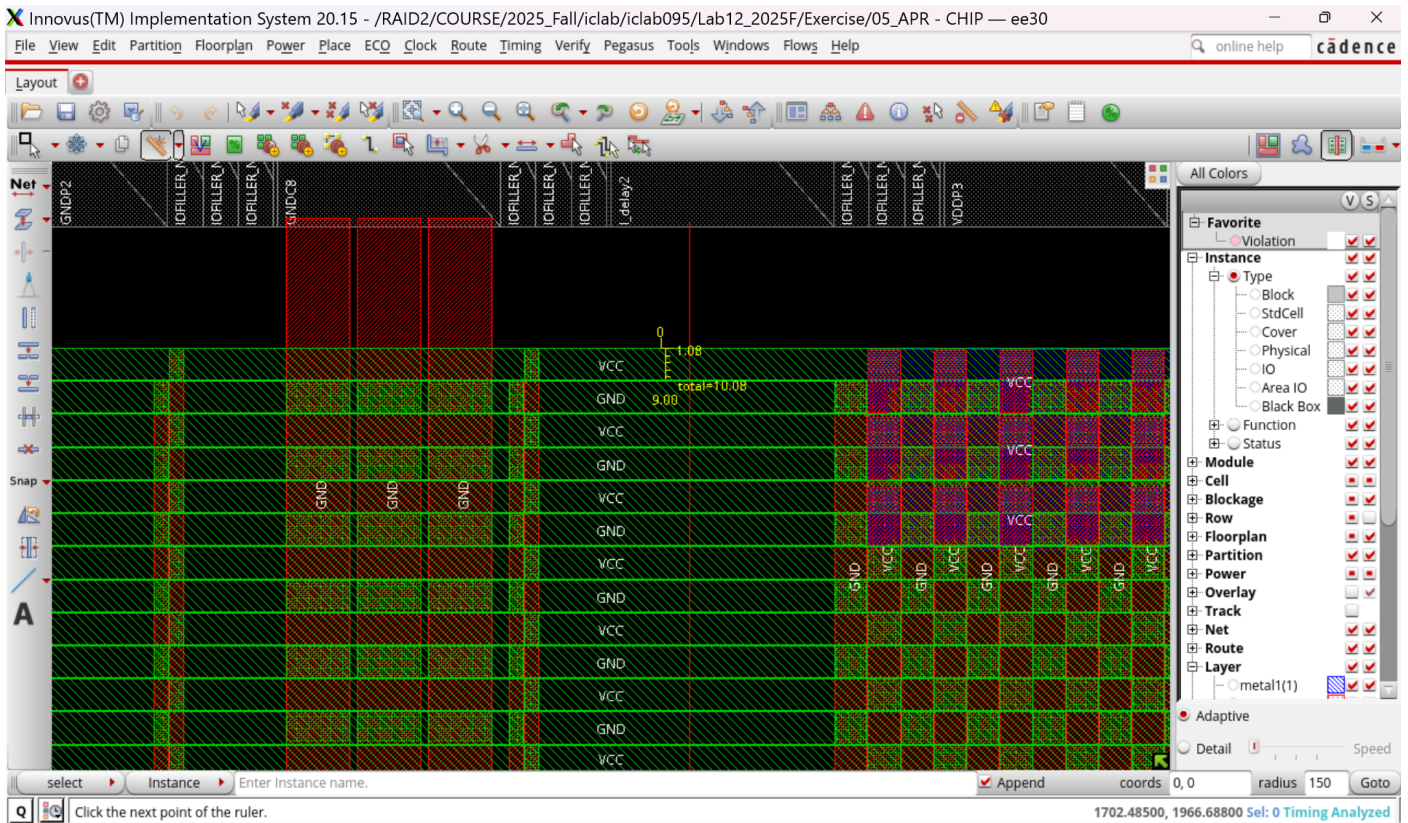
Overall Power Ring



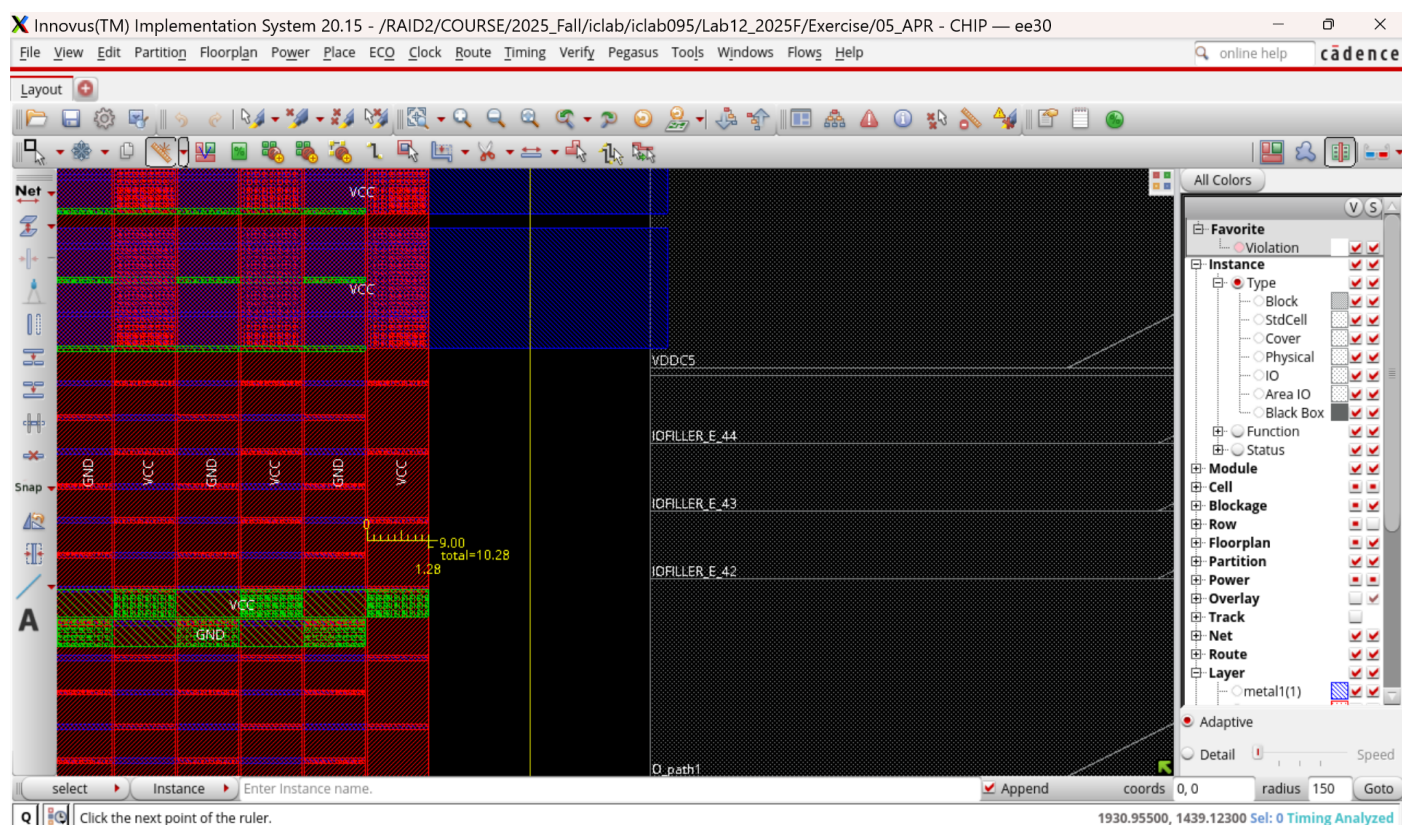
Overall Power Ring: wire group, interleaving, 10 pairs, width 9.



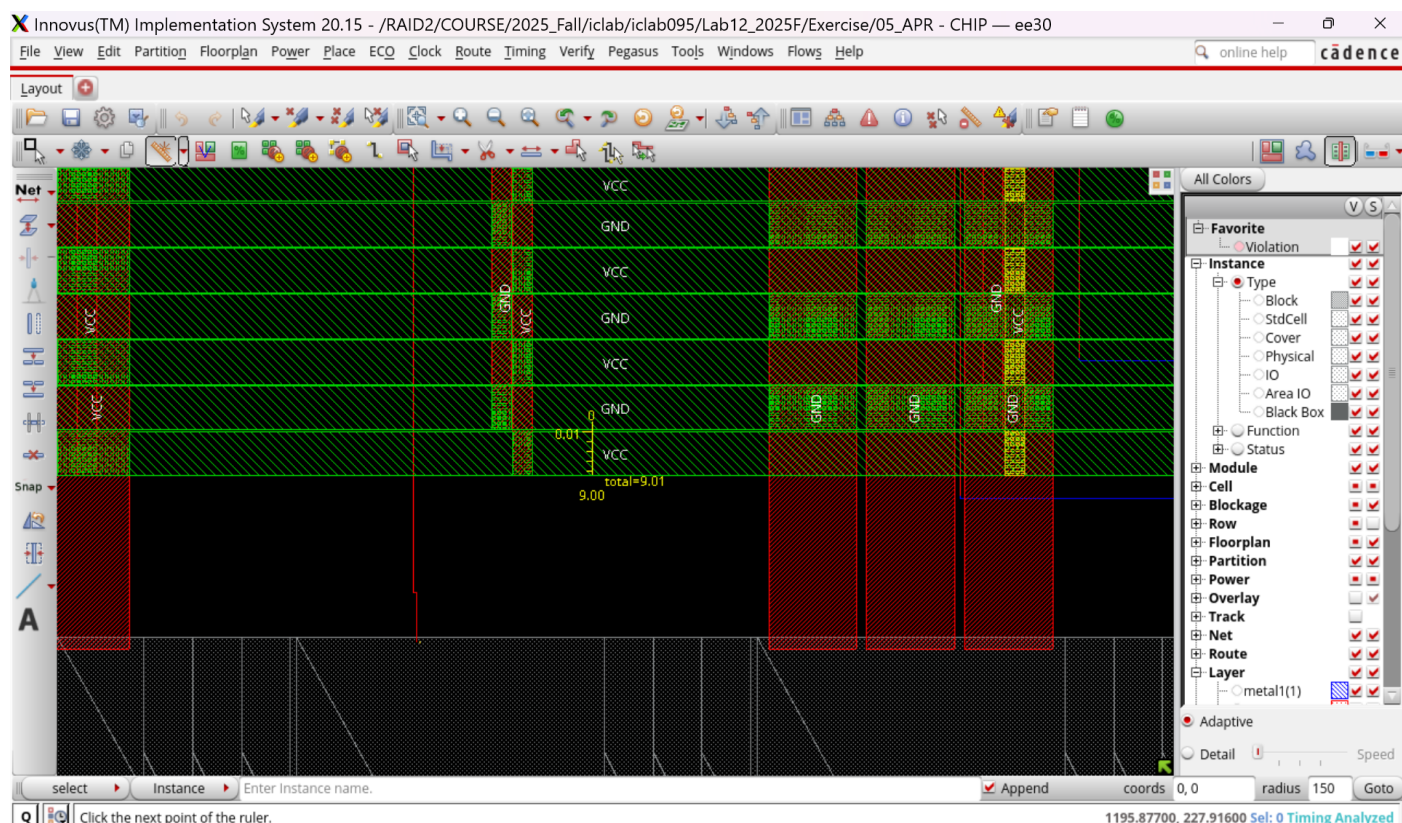
Top Power Ring: width=9



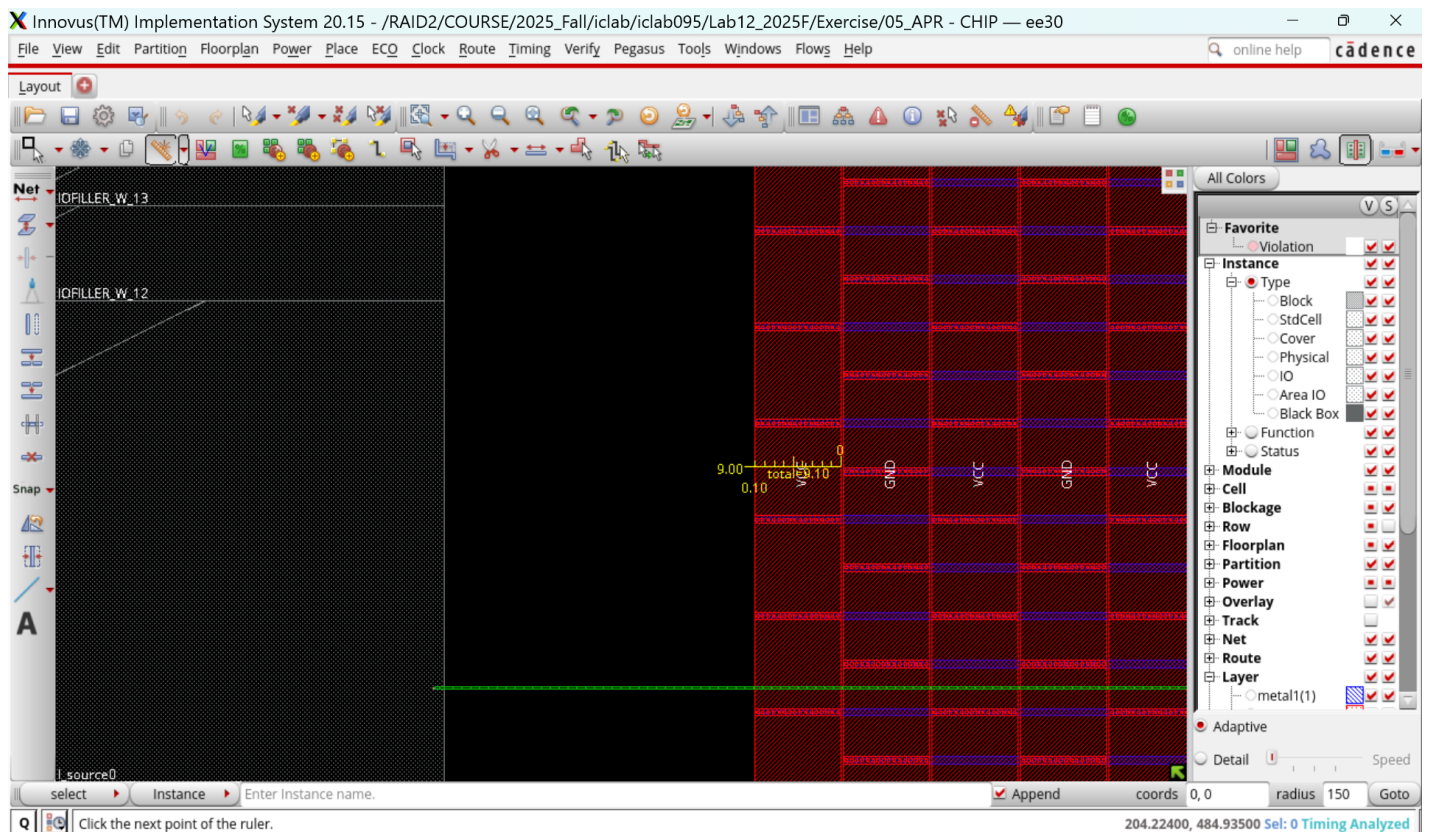
Right Power Ring: width=9



Bottom Power Ring: width=9



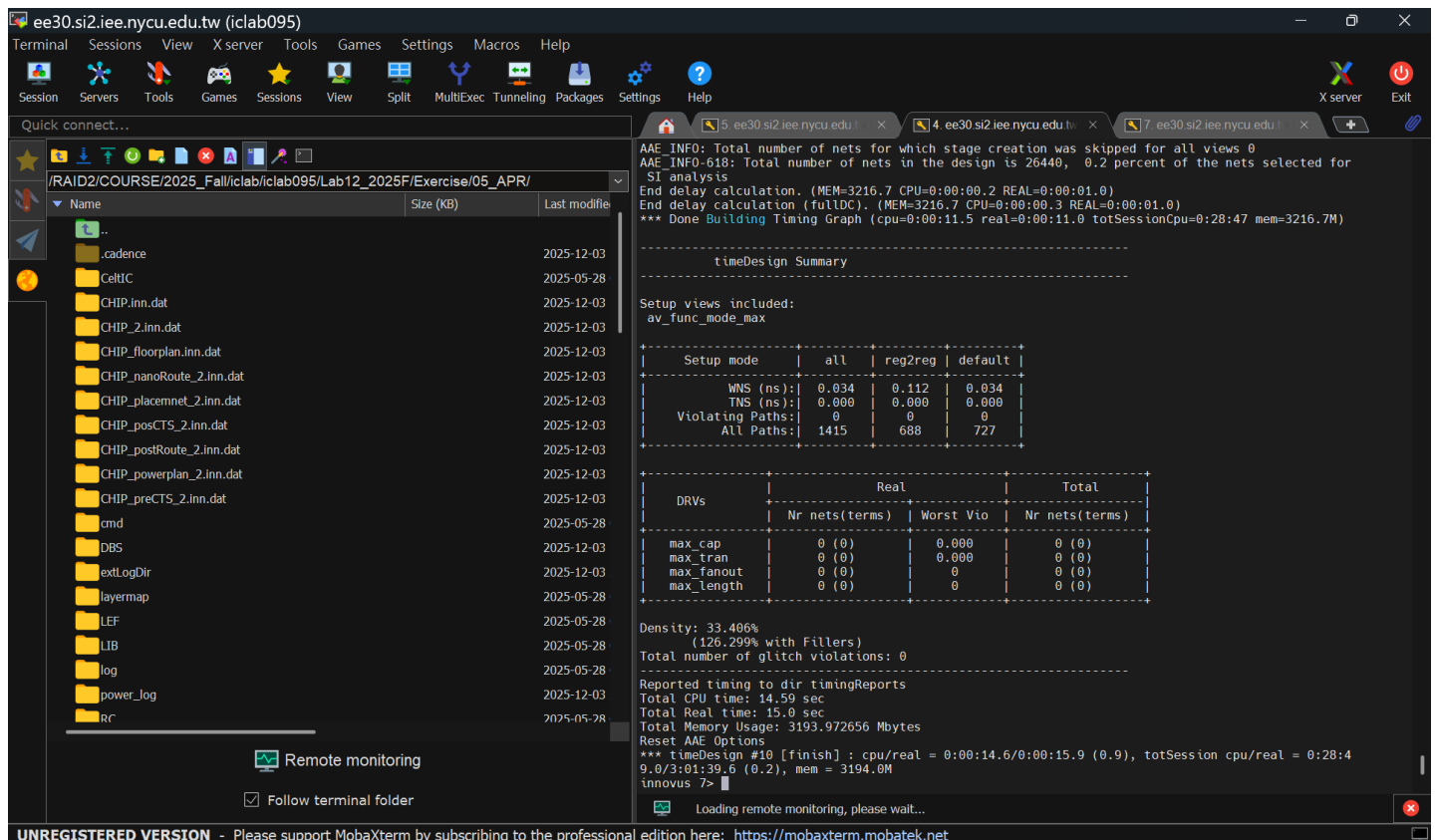
Left Power Ring: width=9



(Width settings are all 9)

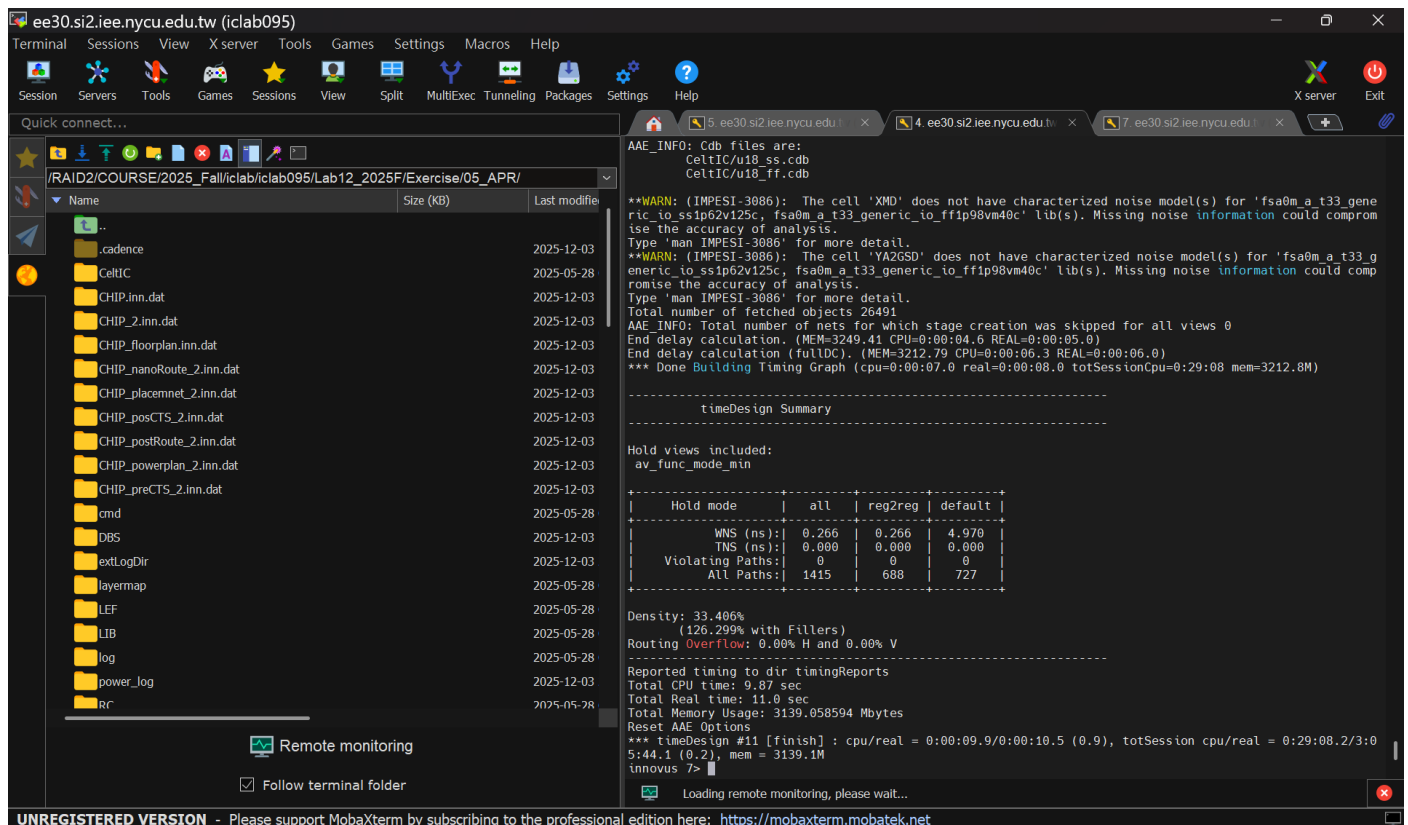
4. Post-Route setup time analysis :

Instructions: After completing the analysis, capture a screenshot of the results displayed in the terminal (including the timeDesign Summary).



5. Post-Route hold time analysis

Instructions: After completing the analysis, capture a screenshot of the results displayed in the terminal (including the timeDesign Summary).



The screenshot shows the MobaXterm interface with a terminal window displaying the results of a post-route hold time analysis. The terminal output includes the following sections:

```
AAE_INFO: Cdb files are:
CeltIC/u18 ss.cdb
CeltIC/u18 ff.cdb

**WARN: (IMPESI-3086): The cell 'XMD' does not have characterized noise model(s) for 'fsa0m_a_t33_gene
ric_to_ssip62v125c, fsa0m_a_t33_generic_to_ffip98vm40c' lib(s). Missing noise information could comprom
ise the accuracy of analysis.
Type 'man IMPESI-3086' for more detail.
**WARN: (IMPESI-3086): The cell 'YA2GSD' does not have characterized noise model(s) for 'fsa0m_a_t33_g
eneric_to_ssip62v125c, fsa0m_a_t33_generic_to_ffip98vm40c' lib(s). Missing noise information could comp
romise the accuracy of analysis.
Type 'man IMPESI-3086' for more detail.
Total number of fetched objects 26491
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=3249.41 CPU=0:00:04.6 REAL=0:00:05.0)
End delay calculation (fullDC). (MEM=3212.79 CPU=0:00:06.3 REAL=0:00:06.0)
*** Done Building Timing Graph (cpu=0:00:07.0 real=0:00:08.0 totSessionCpu=0:29:08 mem=3212.8M)

-----
timeDesign Summary
-----

Hold views included:
av_func_mode_min

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns)  | 0.266 | 0.266 | 4.970 |
| TNS (ns)  | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
+-----+-----+-----+-----+

Density: 33.405%
(126.299% with Fillers)
Routing Overflow: 0.00% H and 0.00% V

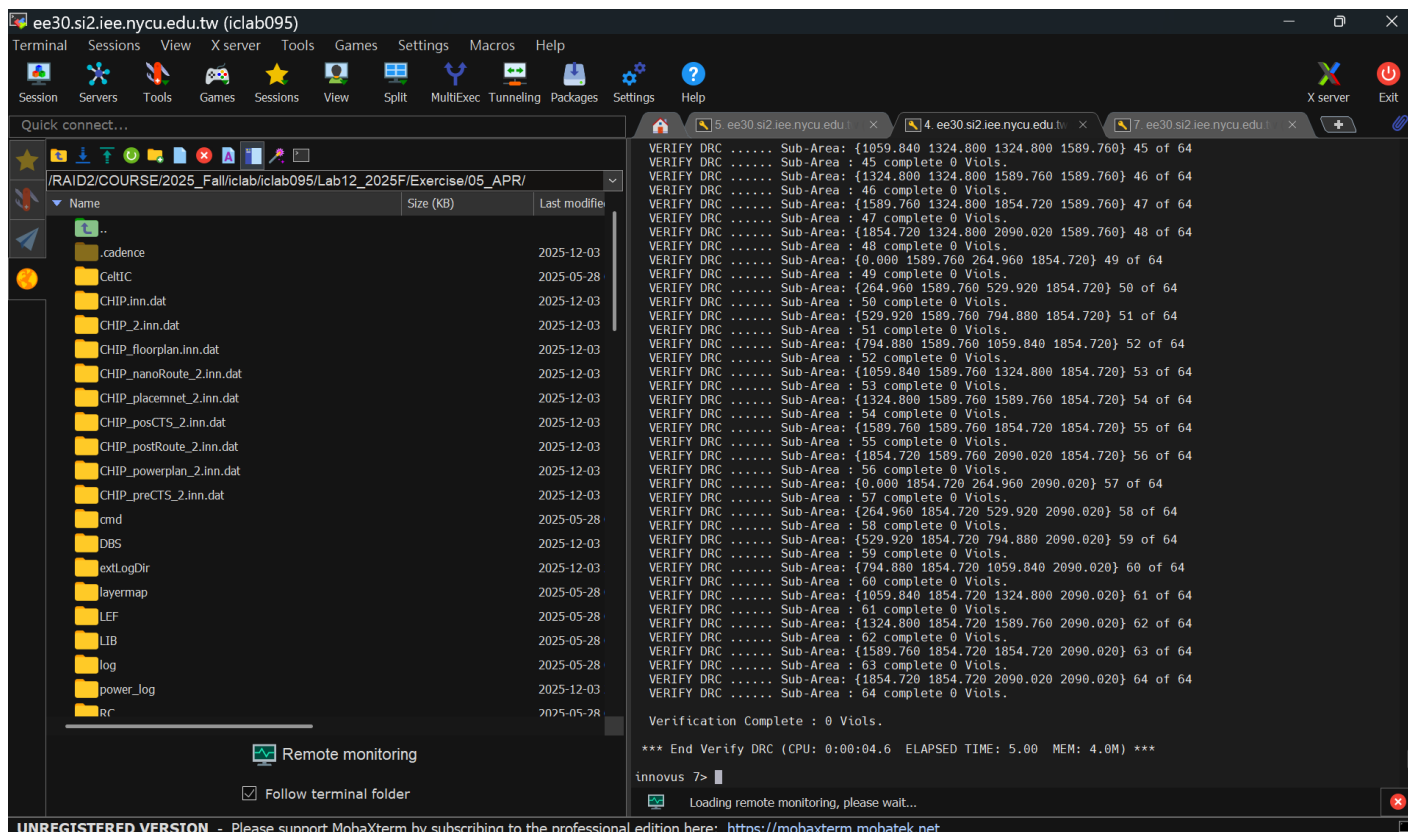
-----
Reported timing to dir timingReports
Total CPU time: 9.87 sec
Total Real time: 11.0 sec
Total Memory Usage: 3139.058594 Mbytes
Reset AAE Options
*** timeDesign #11 [finish] : cpu/real = 0:00:09.0/0:00:10.5 (0.9), totSession cpu/real = 0:29:08.2/3:0
5:44.1 (0.2), mem = 3139.1M
innovus 7>

Loading remote monitoring, please wait...
```

At the bottom of the terminal window, there is a message: "UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>".

6. DRC result

Instructions: After completing the final layout, run the analysis and capture a screenshot of the results displayed in the terminal (including the number of violations).



The screenshot shows the MobaXterm interface with a terminal window displaying the results of a Design Rule Check (DRC). The terminal output includes the following sections:

```
VERIFY DRC ..... Sub-Area: {1059.840 1324.800 1324.800 1589.760} 45 of 64
VERIFY DRC ..... Sub-Area: 45 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1324.800 1589.760 1589.760} 46 of 64
VERIFY DRC ..... Sub-Area: 46 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1324.800 1854.720 1589.760} 47 of 64
VERIFY DRC ..... Sub-Area: 47 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1324.800 2090.020 1589.760} 48 of 64
VERIFY DRC ..... Sub-Area: 48 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1589.760 264.960 1854.720} 49 of 64
VERIFY DRC ..... Sub-Area: 49 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {264.960 1589.760 529.920 1854.720} 50 of 64
VERIFY DRC ..... Sub-Area: 50 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {529.920 1589.760 794.880 1854.720} 51 of 64
VERIFY DRC ..... Sub-Area: 51 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {794.880 1589.760 1059.840 1854.720} 52 of 64
VERIFY DRC ..... Sub-Area: 52 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1059.840 1589.760 1324.800 1854.720} 53 of 64
VERIFY DRC ..... Sub-Area: 53 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1589.760 1589.760 1854.720} 54 of 64
VERIFY DRC ..... Sub-Area: 54 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1589.760 1854.720 1854.720} 55 of 64
VERIFY DRC ..... Sub-Area: 55 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1589.760 2090.020 1854.720} 56 of 64
VERIFY DRC ..... Sub-Area: 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1854.720 264.960 2090.020} 57 of 64
VERIFY DRC ..... Sub-Area: 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {264.960 1854.720 529.920 2090.020} 58 of 64
VERIFY DRC ..... Sub-Area: 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {529.920 1854.720 794.880 2090.020} 59 of 64
VERIFY DRC ..... Sub-Area: 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {794.880 1854.720 1059.840 2090.020} 60 of 64
VERIFY DRC ..... Sub-Area: 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1059.840 1854.720 1324.800 2090.020} 61 of 64
VERIFY DRC ..... Sub-Area: 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1854.720 1589.760 2090.020} 62 of 64
VERIFY DRC ..... Sub-Area: 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1854.720 1854.720 2090.020} 63 of 64
VERIFY DRC ..... Sub-Area: 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1854.720 2090.020 2090.020} 64 of 64
VERIFY DRC ..... Sub-Area: 64 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:04.6 ELAPSED TIME: 5.00 MEM: 4.0M) ***

innovus 7>
```

At the bottom of the terminal window, there is a message: "UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>".

7. LVS result

Instructions: After completing the final layout, run the analysis and capture a screenshot of the results displayed in the terminal (including the number of violations).

The image shows a MobaXterm window with a dark theme. On the left is a file explorer pane showing a directory tree for the path /RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/05_APR/. The tree includes folders like .cadence, CeltC, and various .dat files, along with files like cmd, DBS, extLogDir, layermap, LEF, LIB, log, power_log, and RC. On the right is a terminal window with three tabs. The active tab shows the output of a 'VERIFY CONNECTIVITY' command. The output includes a status bar at the top, a command prompt 'innovus >', and a detailed report of the verification process, including design name, units, boundary, error/warning limits, and a list of processed nets. The report concludes with 'Begin Summary', 'Found no problems or warnings.', 'End Summary', and timing information. At the bottom of the terminal, there is a status bar indicating 'Loading remote monitoring, please wait...'. The MobaXterm window has a top menu bar with options like Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. A bottom status bar indicates 'UNREGISTERED VERSION' and provides a link to the professional edition.

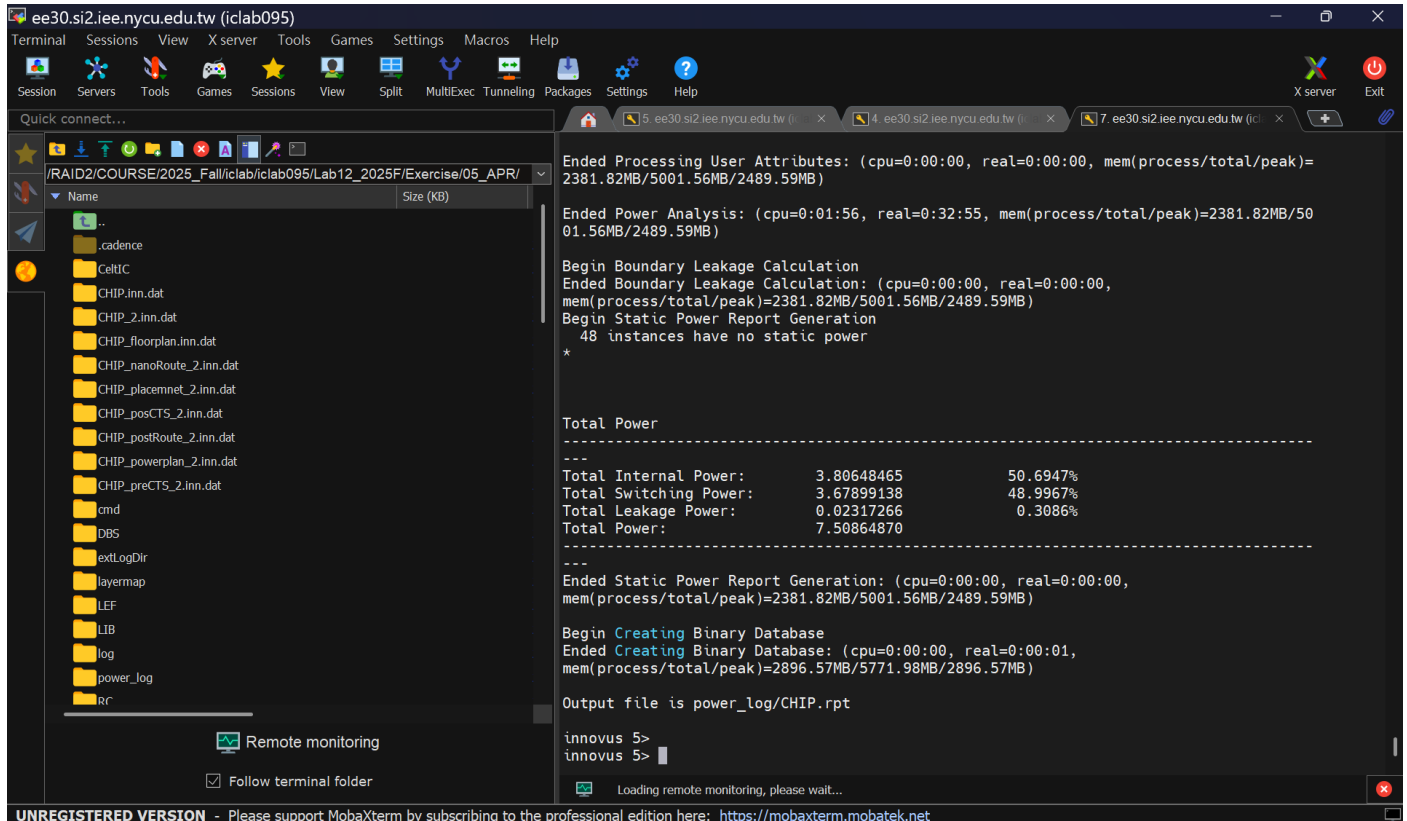
8. Post Layout simulation result

Instructions: After completing the final layout, run the post-simulation in 06_POST and capture a screenshot of the results displayed in the terminal

The image shows a screenshot of the MobaXterm application. The top bar displays the title "ee30.si2.iee.nycu.edu.tw (iclab095)". Below the title bar is a menu bar with options: Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. A toolbar with icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, and Help is located below the menu bar. The main window is divided into three panes. The left pane shows a file explorer with the path "/RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/06_POST/". It lists files and folders including "csrc", "slmv.daidir", "01_run_vcs_post", "02_irun_post", "03_xrun_post", "04_verdi", "05_nWave", "08_check", "09_clean_up", "CHIP.sdf", "CHIP.v", "CHIP_POST.fsd", "filelist.f", "makefile", "novas_dump.log", "PATTERN.v", "slmv", "TESTBED.v", "udl.key", and "vcs.log". The right pane shows the terminal output, which includes a congratulatory message and simulation results: "Congratulations! execution cycles = 50000 clock period = 11.000000ns". It also shows the command "\$finish called from file 'PATTERN.v', line 118." and the simulation time "1003228500". The bottom status bar indicates "UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net".

9. Power result

After running the analysis, capture a screenshot of the results in the terminal (including Total Power).



The terminal window displays the following output:

```
Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2381.82MB/5001.56MB/2489.59MB)

Ended Power Analysis: (cpu=0:01:56, real=0:03:55, mem(process/total/peak)=2381.82MB/5001.56MB/2489.59MB)

Begin Boundary Leakage Calculation
Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2381.82MB/5001.56MB/2489.59MB)
Begin Static Power Report Generation
48 instances have no static power
*

Total Power
-----
Total Internal Power:      3.80648465      50.6947%
Total Switching Power:    3.67899138      48.9967%
Total Leakage Power:      0.02317266      0.3086%
Total Power:              7.50864870

-----
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2381.82MB/5001.56MB/2489.59MB)

Begin Creating Binary Database
Ended Creating Binary Database: (cpu=0:00:00, real=0:00:01, mem(process/total/peak)=2896.57MB/5771.98MB/2896.57MB)

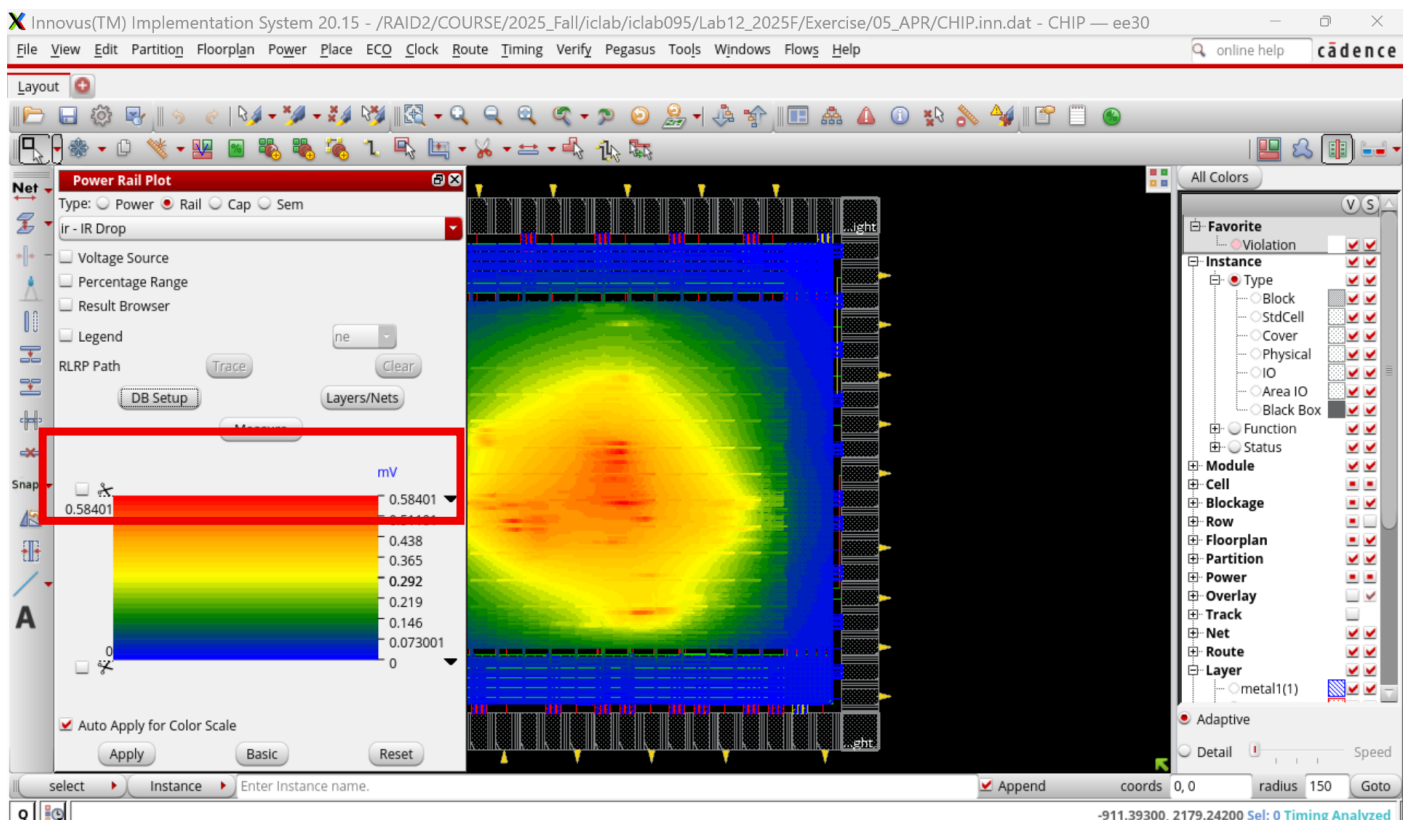
Output file is power_log/CHIP.rpt

innovus 5>
innovus 5>
```

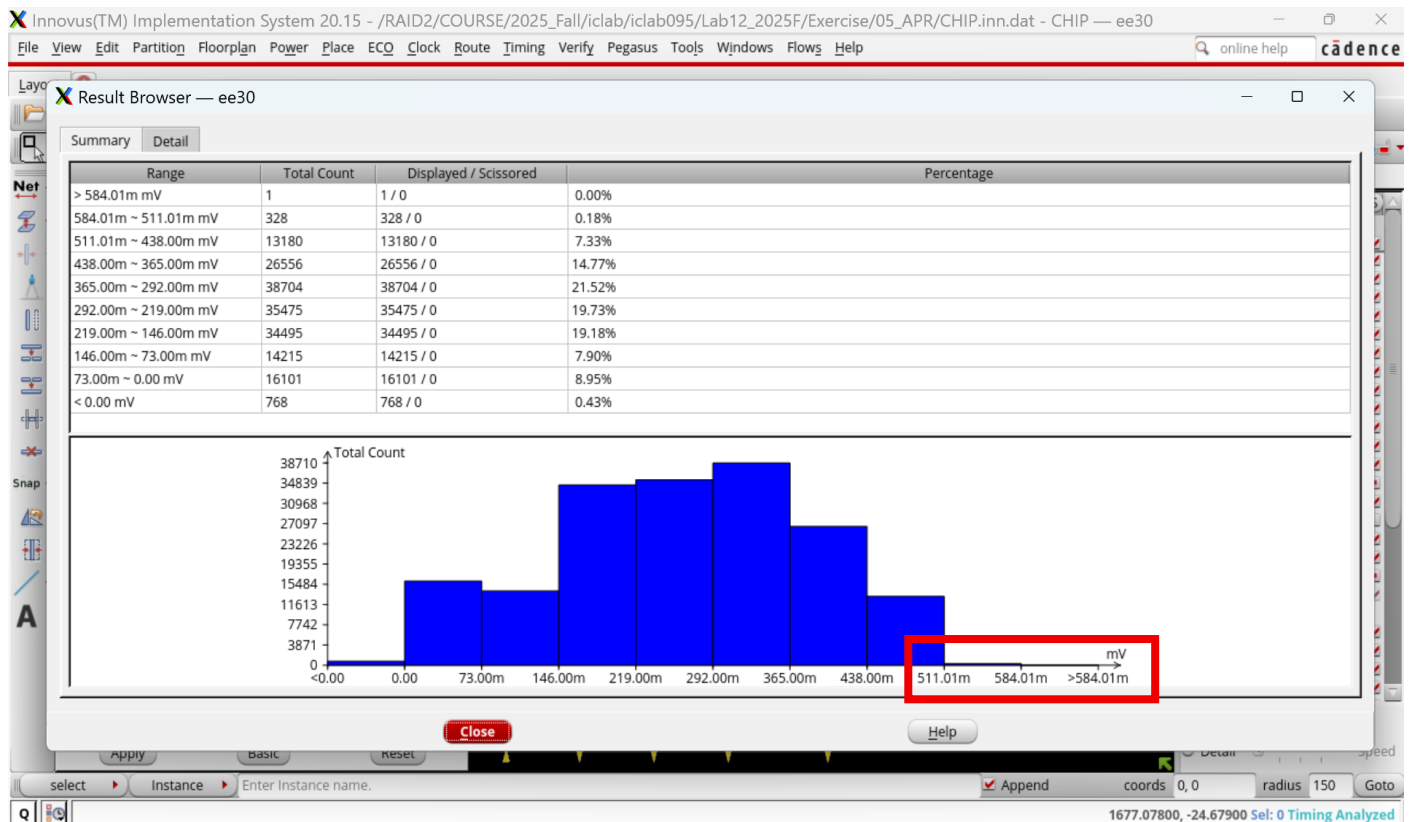
10. IR Drop Results :

Instructions: After completing the analysis, capture a screenshot of the results in Innovus. **Highlight the maximum value of the IR drop** and **provide a brief explanation of the method you used** to mitigate the IR drop issue.

maximum value of the IR drop: 0.58401mV



maximum value of the IR drop: 0.58401mV



Explanation:

目前的作法是依照助教的建議，增加 Core PowerPad 與 IO PowerPad 的數量。其中，IO PowerPad 共配置 8 組，而 Core PowerPad 則配置 16 組。

在 PowerPad 的擺放上，我也盡量依助教所述，避免將 PowerPad 集中於晶片的單一邊，而是盡可能均勻分佈於四周，以避免造成某些區域的邏輯閘因電源距離過遠而產生不利影響。

此外，目前的 power stripe 寬度設定為 4。本來有考慮將 stripe 與 stripe 之間間距（原為 100）再縮小，但由於目前的 IR Drop 檢查已通過，因此暫時尚未調整。

若後續需要再優化，則會考慮進一步增加 PowerPad 數量，或縮小 stripe 間距，不過此部分仍需視 DRC 是否能通過而定。